

CPD 2023: The 6th International Workshop on Combining Physical and Data-Driven Knowledge in Ubiquitous Computing

Yuanzhang Xiao
University of Hawaii at Manoa
Honolulu, HI, United States
yxiao8@hawaii.edu

Jie Xu
University of Miami
Miami, FL, United States
jiexu@miami.edu

Weiwei Jiang
Beijing University of Posts and
Telecommunications
Beijing, China
jwwthu@gmail.com

ABSTRACT

With the proliferation of connected devices with advanced sensing, computing, and communication capabilities, ubiquitous computing systems have become prevalent nowadays. They have the potential to revolutionize various industries by enabling new applications and services (e.g., patient monitoring, personalized recommendations, traffic control, home energy management). However, in real-world ubiquitous computing systems, data collection can be expensive or impossible. Due to the limited quantity and quality of data available, pure data-driven methods may not perform well. A promising approach to overcome these limitations is to utilize physical knowledge, including domain knowledge from experts, heuristics based on experience, and analytic models of physical phenomena.

The theme of this workshop is to advance the theoretical understanding, algorithmic development and system implementations of ubiquitous computing systems that integrate physical knowledge with data-driven methods. The workshop will provide an inclusive gathering for researchers and practitioners from various fields and facilitate future collaborations. The workshop welcomes research papers as well as position papers. The overall goal is to grow the community who are dedicated to improving ubiquitous computing systems by fusing physical knowledge into data-driven methods.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing.

KEYWORDS

ubiquitous computing, data-driven, physical knowledge, machine learning, artificial intelligence, cyber-physical system.

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1 TOPIC, RATIONALE, AND OBJECTIVES

The workshop aims to stimulate the research and discussions on **systematically utilizing physical knowledge to enhance pure data-driven methods in ubiquitous computing**.

A key challenge in ubiquitous computing systems is the lack of control in data quality. The data collected may be noisy, unlabeled, and non-stationary. These limitations may adversely affect the performance of pure data-driven methods.

In response to these challenges, researchers are increasingly recognizing the benefits of incorporating physical knowledge into data-driven methods. Here, the physical knowledge can be domain knowledge from experts, heuristics based on experience, and analytic models of the underlying physical processes. By integrating this knowledge into data-driven methods, researchers can improve the accuracy and robustness of these methods in ubiquitous computing systems.

By bringing together researchers from different disciplines and applications, we strive to strengthen the understanding at the interface of physical knowledge based and data-driven methods, discover new problems, and foster interdisciplinary collaborations.

2 RELEVANCE AND TIMELINESS

The ubiquitous computing community mainly focuses on human-centric technologies, where the data is multimodal and collected from heterogeneous sensing systems at the societal and personal levels. As this field of research grows, an increasing number of papers address the limitations of pure data-driven approaches.

To date, there is a lack of systematic approaches of combining physical and data-driven knowledge. With the growth of the systems research and machine learning, we hope this workshop can grow this new sub-field of computing that combines physical knowledge and data-driven methods.

UbiComp gathers researchers that work on various ubiquitous computing applications, with an emphasis on design, development, and deployment. Many applications that aim to understand the human and environmental aspects of the physical world face challenges in obtaining sufficient labeled data to accurately infer information. These challenges underscore the importance of exploring novel approaches, such as combining physical knowledge with data-driven methods, which will be a focus of our workshop. Therefore, both the topics and the interdisciplinary theme of our workshop fit the UbiComp conference very well.

Previous CPD workshops. The previous five workshops at UbiComp 2018, 2019, 2020, 2021, and 2022 attracted papers working on the limitation of pure data-driven approaches appearing in different aspects. In the past five years, we received more than 100 submissions from Asia, Australia, Europe and North America, which shows the worldwide impact of the workshop. In addition, we attracted more than 50 audience to attend our workshop last year. Based on the historical numbers of submissions and attendees, we expect a growth of the workshop this year.

The topics/sessions we cover last year include

- Innovations in learning algorithms that combine physical knowledge or models for sensor perception and understanding
- Mobile and pervasive systems that utilize physical knowledge to enhance data acquisition
- Heterogeneous collaborative sensing based on physical rules
- Distributed sensing for cyber-physical systems
- Human centric learning

Each topic drew a lot of attentions and participants had enthusiastic discussions over these topics.

3 PROPOSED ORGANIZERS AND PROGRAM COMMITTEE

The organization consists of three workshop chairs and three TPC chairs, an advising committee, and several program committee members. We list proposed personnel and their functions here.

3.1 Workshop Chairs

- Yuanzhang Xiao (University of Hawaii at Manoa)
- Jie Xu (University of Miami)

3.2 Workshop TPC Chairs

- Weiwei Jiang (Beijing University of Posts and Telecommunications)
- Amir H. Alavi (University of Pittsburgh)

3.3 Workshop Web Chair

- Letian Zhang (Middle Tennessee State University)

3.4 Tentative Program Committee

- Yaxiong Xie (Princeton University)
- Susu Xu (Stony Brook University)
- Di Jin (Amazon Alexa AI)
- Weixi Gu (China Academy of Industrial Internet)
- Shuai Wang (Southeast University)
- Jason Shuo Zhang (University of Colorado Boulder)
- Weitao Xu (City University of Hongkong)
- Huandong Wang (Massachusetts Institute of Technology)
- Fengli Xu (University of Chicago)
- Wan Du (University of California Merced)
- Shijia Pan (University of California Merced)
- Mostafa Mirshekari (Stanford University)
- Chris Xiaoxuan Lu (Liverpool University)
- Miadreza Shafie-khah (University of Vaasa)
- Shahab Bahrani (The University of British Columbia)

- Marzieh Khakifirooz (Tecnológico de Monterrey, Mexico)
- M. Hadi Amini (Florida International University)
- Yong Zhuang (Carnegie Mellon University)
- Dezhi Hong (University of California San Diego)
- Roozbeh Jafari (Texas A&M University)
- Jun Han (National University of Singapore)
- Mi Zhang (Michigan State University)
- Pan Hu (Stanford University)
- Miadreza Shafie-khah (University of Vaasa)
- Shahab Bahrani (The University of British Columbia)
- Marzieh Khakifirooz (Tecnológico de Monterrey, Mexico)
- Ishara Dharmasena (Loughborough University, UK)
- Sicong Liu (Xiamen University)
- Amir H. Alavi (University of Pittsburgh)
- Yongpan Zou (Shenzhen University)
- Linqi Song (City University of Hong Kong)
- Wenbo Ding (Tsinghua University)
- Zhengxiong Li (University of Colorado Denver)
- Xinlei Chen (Tsinghua University)
- Le Yang (Xi'an Jiaotong University)

The Program Committee members will conduct following tasks with the Workshop chairs:

- reviewing submitted papers (see Planned Review Procedure)
- participate review discussion if needed
- make decisions on the paper acceptance/rejection

3.5 Advising Committee

The Advising Committee members are experienced conference organizers that can provide feedback and guidance to workshop chairs.

- Linqi Song (City University of Hong Kong)
- Wenbo Ding (Tsinghua University)
- Xinlei Chen (Tsinghua University)
- Shijia Pan (University of California Merced)

3.6 Important Dates

- Submission Deadline: May 26, 2023
- Author Notifications: June 30, 2023
- Camera-ready Deadline: July 31, 2023
- Workshop: October 8, 2023

4 PLANNED REVIEW PROCEDURE

Each paper will be reviewed by at least three members. The criteria for accepted papers are consistent with our previous workshops, including

- if the paper address or reveal the limited data problem in pure data-driven methods;
- if the physical knowledge is used to enable better information processing;
- if the paper/solution/observation is able to trigger in-depth discussions regarding the combination of physical and data-driven knowledge.

5 TARGET ATTENDEE AND EXPECTED NUMBERS

We target at original work on

- solving a particular problem caused by the limitation of the data through the physical knowledge of the limitation
- observation on physical phenomena that causes data limitations/inference issues in multiple applications
- introducing emerging data-driven techniques at the intersection of ubiquitous computing and machine learning

The former allows the audience to broaden their understanding of the physical knowledge that can affect inference from limited data. The later inspires more generalized approaches to be further explored. Since the limitation of the data mostly appeared in sensing data from physical systems, our target authors are mainly from (not limited to) the community of UbiComp, CPS-IoT Week, and SenSys/BuildSys . The expected submission is 20 to 30, and the target number of papers acceptance is 12 to 20. We will host four

sessions (two in the morning and two in the afternoon), and each session contains three presentations of the accepted papers. There will be a 30 mins break between each two consecutive sessions so that the audience can discuss with the author in a timely fashion.

6 PLANNED PUBLICITY PROCEDURE

We will send the call for paper (CFP) to the relevant email lists and our research colleagues' emails. We plan to send out the CFP on the following dates 04/15/2023, 04/30/2023, 05/15/2023. We also plan to post the workshop information on social media including LinkedIn, Twitter, Facebook, and WeChat.

7 PLANNED WORKSHOP ACTIVITY

We plan to have invited talks, contributed talks, and a poster session. Invited talks may be a 12-min presentation followed by a 5-min Q&A discussion. Contributed talks will be a 12-min presentation followed by 3-min Q&A discussion.